Laboratory Index

1. Write a program to read and display a digital image, convert it into grayscale, and determine its basic properties such as dimensions, number of channels, and intensity range.
2. Write a program to perform sampling and quantization on an image by reducing its resolution and number of gray levels.
3. Write a program to determine pixel relationships by finding neighbors and Euclidean and Manhattan distance between pixels in an image.
4. Write a program to perform point processing operations: image negative, log transformation, contrast stretching, and thresholding.
5. Write a program to perform bit plane slicing.
6. Write a program to compute, display, and perform histogram equalization on an image.
7. Write a program to apply mean and median smoothing filters on an image and display the results.
8. Write a program to apply image sharpening on an image using Laplacian and Sobel filters and display the original and sharpened results.
9. Write a program to compute and visualize the Fast Fourier Transform of an image and apply frequency domain filtering.
10. Write a program to add Gaussian noise and salt-and-pepper noise to a digital image and restore it using appropriate filtering techniques
11. Write a program to perform image compression using technique using Huffman Coding.
12. Write a program to perform morphological operations such as dilation, erosion, opening, and closing on binary images.
13. Write a program to perform line and edge detection on a given grayscale image using appropriate masks and display the results for each technique.